

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

13

10 answers · Rationale-rich · Teach from doc

Q1**D****D.** $55x^2$

Choice D is correct. The given expression shows addition of two like terms. Therefore, the given expression is equivalent to $50 + 5x^2$, or $55x^2$.

Choice A is incorrect. This expression is equivalent to $505x^2$, not $50 + 5x^2$.

Choice B is incorrect. This expression is equivalent to $\frac{50}{5}x^2$, not $50 + 5x^2$.

Choice C is incorrect. This expression is equivalent to $50 - 5x^2$, not $50 + 5x^2$.

Q2**3**

The correct answer is 3. For the graph of the exponential function f shown, where $y = fx$, it's given that the y -intercept of the graph is 0, y . The graph intersects the y -axis at the point 0, 3. Therefore, the value of y is 3.

Q3**D****D.** $j = \frac{k + 15m}{8}$

Choice D is correct. To express j in terms of k and m , the given equation must be solved for j . Dividing each side of the given equation by 8 yields $j = \frac{k + 15m}{8}$.

Choice A is incorrect. This is equivalent to $8j = k + 120m$.

Choice B is incorrect. This is equivalent to $8j = 8k + 15m$.

Choice C is incorrect. This is equivalent to $\frac{j}{8} = k + 15m$.

Q4**C****C.** 18

Choice C is correct. The value of $f(2)$ is found by evaluating the expression $2(3^x)$ when $x = 2$. Substituting 2 for x in the given equation yields $f(2) = 2(3^2)$. Simplifying 3^2 in the equation results in $f(2) = 2(9)$. Evaluating the right-hand side of the equation yields $f(2) = 18$. Therefore, the value of $f(2)$ is 18.

Choice A is incorrect and may result from evaluating the expression as (3^2) . Choice B is incorrect and may result from evaluating the expression as $2(3 \cdot 2)$. Choice D is incorrect and may result from evaluating the expression as $(2 \cdot 3)^2$.

Q5**B****B.** $p = q + r - 34$

Choice B is correct. Subtracting 34 from each side of the given equation yields $p = q + r - 34$. Thus, the equation $p = q + r - 34$ correctly expresses p in terms of q and r .

Choice A is incorrect. This equation can be rewritten as $p - 34 = q + r$.

Choice C is incorrect. This equation can be rewritten as $p - 34 = -q - r$.

Choice D is incorrect. This equation can be rewritten as $p + 34 = -q - r$.

Q6**B**

B. $h(n) = 22(0.85)^n$

Choice B is correct. It's given that for the function h , hn is the estimated maximum height, in feet, of the ball after it has hit the ground n times. It's also given that the function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. It follows that h is a decreasing exponential function that can be written in the form $hn = a\frac{p}{100}^n$, where a is the initial height, in feet, the ball was dropped from and the function estimates that the maximum height reached after each time the ball hits the ground is $p\%$ of the maximum height reached after the previous time the ball hit the ground. It's given that the ball is dropped from an initial height of 22 feet. Therefore, $a = 22$. Since the function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground, $p = 85$. Substituting 22 for a and 85 for p in the equation $hn = a\frac{p}{100}^n$ yields $hn = 22\frac{85}{100}^n$, or $hn = 22(0.85)^n$.

Choice A is incorrect. This function estimates that the maximum height reached after each time the ball hits the ground is 22%, not 85%, of the maximum height reached after the previous time the ball hit the ground.

Choice C is incorrect. This function estimates that the ball is dropped from an initial height of 85 feet, not 22 feet, and that the maximum height reached after each time the ball hits the ground is 22%, not 85%, of the maximum height reached after the previous time the ball hit the ground.

Choice D is incorrect. This function estimates that the ball is dropped from an initial height of 85 feet, not 22 feet.

Q7**C****C.** 1,250

Choice C is correct. Each point (x, y) on the graph represents the elevation x , in meters, and the corresponding measure of plant diversity y in a region of Switzerland. Therefore, the point on the graph with the greatest y -coordinate represents the location that has the greatest measure of plant diversity in the region. The greatest y -coordinate of any point on the graph is approximately 13,500. The x -coordinate of that point is approximately 1,250. Therefore, the closest elevation at which the plant diversity is the greatest is 1,250 meters.

Choice A is incorrect. This value is closest to the greatest y -coordinate of any point on the graph and therefore represents the greatest measure of plant diversity, not the elevation where the greatest measure of plant diversity occurs. Choice B is incorrect. At an elevation of 3,000 meters the measure of plant diversity is approximately 4,000. Because there are points on the graph with greater y -coordinates, 4,000 can't be the greatest measure of plant diversity, and 3,000 meters isn't the elevation at which the greatest measure of plant diversity occurs. Choice D is incorrect. At an elevation of 250 meters, the measure of plant diversity is approximately 11,000. Because there are points on the graph with greater y -coordinates, 11,000 can't be the greatest measure of plant diversity and 250 meters isn't the elevation at which the greatest measure of plant diversity occurs.

Q8**C****C.** 0, 2

Choice C is correct. The vertex of the graph of a quadratic function in the xy -plane is the point at which the graph is either at its minimum or maximum y -value. In the graph shown, the minimum y -value occurs at the point 0, 2.

Choice A is incorrect. The graph shown doesn't pass through the point 0, - 2.

Choice B is incorrect. The graph shown doesn't pass through the point 0, - 3.

Choice D is incorrect. The graph shown doesn't pass through the point 0, 3.

Q9**7**

The correct answer is 7. It's given that the parabola intersects the y -axis at the point x, y . The graph shows that the parabola intersects the y -axis at the point $0, 7$. Therefore, the value of y is 7.

Q10**C****C.** 309

Choice C is correct. It's given that function g is defined by $g(x) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g(81) = \sqrt{81} + 300$, which is equivalent to $g(81) = 9 + 300$, or $g(81) = 309$. Therefore, the value of $g(x)$ when $x = 81$ is 309.

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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